

Gesture Recognition on Smartphones

It is time we used those front-facing cameras on smartphones for more than just video-chatting, 'thank you' messages, clicking pictures for self portraits and other routine tasks. Are smartphones ready for Kinect-like gesture control? Let's find out

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There are thousands of cellphone models available in the market, all with varying features. In recent years, cellphones have also seen considerable progress in terms of user interaction, evolving all the way from QWERTY keypads to touchscreens. What could be the next transition in user interface?

The evolution of user interfaces

Cellphones have come a long way since their introduction, both in function as well as methods to use and control these devices, which has always been a challenge.

The journey started with those bulky phones that incorporated long antennae, physical switches and buttons; for example the Analog Motorola Dyna 8000X. The next round of innovation pushed touch-sensitive interaction with trackpads, such as in the IBM Simon. Currently, touchscreen is the most dominating and familiar interface for cellphone users around the world. Simultaneously, work has also been done to enable voice-interaction with these electronic devices. This aspect too has come a long way but is still not robust enough to cover all of our needs.

Jumping from touchscreens to gesture control

Over the last two years, we have seen smartphones explode in their capabilities. However, there is still room for improvement in user experience. Touch was the last great user-interface advancement for cellphones and hand gesture recognition could be the next.

Gesture recognition is perceived

Applications of Gesture Recognition in Today's Mobile Phones

A few use-cases that will enhance the mobile phone's user experience are:

1. Answer an incoming call (speaker-on) with a wave of the hand while driving.
2. Skip tracks or control volume on your media player using simple hand motions. Lean back and, with no need to shift to the device, control what you watch/listen to.
3. Scroll webpages or pages within an eBook with simple left- and right-hand gestures. This is ideal when touching the device is a barrier, such as hands are wet, dirty or wearing gloves. (We all are familiar with the annoying smudges on the screen caused by touching.)
4. Use the smartphone as a media hub. The user can dock the device to the TV and watch content from the device while controlling the content in a touch-free manner from afar.

today as the natural evolution of intuitive user interfaces. Since the creation of touchscreens, gestures have reigned in an entirely new aspect as to how we interact with our devices. Gestures allow users to perform specific tasks in an extremely efficient and dynamic manner.

Some of the most used gestures are swipe to unlock, pinch to zoom and pull to refresh. While these are relatively basic by most means, gestures have

concept to Microsoft's Xbox Kinect but on a much smaller scale.

Adding a whole new dimension to multimedia

Touch-free functionality in smartphones has added a brand new dimension to mobile games and applications. We can now take advantage of the Z-axis by using simple hand shapes to enable reality-like experiences.



Fig. 1: Evolution of mobile phones (Courtesy: wordpress.com)

evolved greatly. There are a number of different companies trying to push forward in touch-free gesture control development. Third-party developers have begun to truly utilise the potential of touch-free control, all within their apps. Gesture recognition can offer an intuitive way to interact with a mobile phone. It is almost similar in



Fig. 2: The Qualcomm QCP-2700, a mid-1990s candybar style phone, and iPhone 5, a current production smartphone (Courtesy: Wikipedia)

Actions supported in games include bowling, throwing darts, playing rock-paper-scissors and more.

Other scenarios made easier with hand gesture recognition are controlling a phone with dirty or wet hands, managing calls while driving or working, and interacting with a smartphone when wearing